

KIMCHAEK SONGJIN, Korea, Democratic People's Republic of

WMO: 470250

Lat: 40.667N

Lon: 129.200E

Elev: 75

StdP: 14.66

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.3	13.4	-13.3	2.7	15.1	-7.9	3.6	18.0	17.8	22.3	15.2	25.2	3.5	320	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	83.7	75.0	80.8	73.8	78.5	72.6	77.3	81.7	75.6	79.3	74.1	77.4	6.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.8	135.3	80.5	74.3	128.6	78.5	72.9	122.6	76.8	40.9	81.8	39.3	79.5	37.9	77.5	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	14.7	11.7	DB	4.2	90.0	3.4	2.8	1.7	91.9	-0.3	93.6	-2.2	95.1	-4.6	97.1	(4)
(5)			WB	2.3	79.8	3.1	2.6	0.1	81.7	-1.8	83.2	-3.5	84.7	-5.8	86.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	25.9	28.4	37.1	47.5	55.3	62.7	70.2	73.3	65.7	55.1	41.7	30.4	(6)	
	DBStd	16.96	5.81	5.97	6.04	5.84	5.75	4.44	5.09	3.76	4.18	5.31	6.99	6.57	(7)	
	HDD50	2768	748	605	403	115	15	0	0	0	0	17	258	607	(8)	
	HDD65	6159	1213	1025	865	524	306	97	11	1	39	308	698	1072	(9)	
	CDD50	2607	0	0	2	41	179	381	627	722	470	175	10	0	(10)	
	CDD65	524	0	0	0	0	5	28	173	258	59	1	0	0	(11)	
	CDH74	2135	0	0	0	5	37	83	667	1173	167	3	0	0	(12)	
	CDH80	347	0	0	0	0	5	14	129	180	19	0	0	0	(13)	
	WSAvg	3.4	3.2	3.6	4.1	4.4	3.7	2.9	2.7	3.0	3.4	3.7	3.4	3.1	(14)	
	PrecAvg	27.7	0.9	0.7	0.7	1.4	2.3	2.8	6.1	5.4	3.5	1.6	1.5	1.0	(15)	
(16) Precipitation	PrecMax	48.8	3.5	2.4	3.4	3.6	5.1	8.7	15.5	14.0	11.3	5.5	4.1	3.0	(16)	
	PrecMin	18.9	0.2	0.0	0.0	0.2	0.2	0.4	0.2	0.6	0.0	0.0	0.0	0.2	(17)	
	PrecStd	7.1	0.7	0.6	0.8	0.9	1.3	2.1	4.1	3.1	2.9	1.2	1.1	0.7	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.1	47.6	61.1	72.5	78.9	81.6	87.7	88.2	82.7	72.8	63.4	50.0	(19)	
		MCWB	38.2	39.6	48.8	55.9	60.5	66.2	76.3	78.4	70.4	61.9	53.2	43.6	(20)	
	2%	DB	39.8	43.0	53.6	65.6	72.8	75.9	83.5	84.2	78.1	69.2	59.2	46.0	(21)	
		MCWB	34.4	37.7	45.2	51.4	58.5	65.0	75.3	76.7	69.0	60.8	51.9	41.2	(22)	
	5%	DB	37.3	40.2	49.3	61.0	68.1	71.7	80.1	81.6	75.3	66.8	56.0	43.1	(23)	
		MCWB	32.6	36.0	42.4	49.6	56.8	65.2	74.4	75.2	67.7	59.2	49.6	38.3	(24)	
	10%	DB	35.0	37.6	46.3	57.2	63.9	69.2	77.5	79.4	73.2	64.6	52.7	40.3	(25)	
		MCWB	31.2	34.0	40.7	48.0	55.4	64.5	73.1	74.0	66.5	58.1	47.1	36.0	(26)	
	0.4%	WB	39.2	41.8	50.1	57.2	63.4	69.5	79.3	80.0	73.9	66.0	55.8	45.1	(27)	
		MCDB	42.7	45.3	58.6	70.0	72.6	73.8	85.2	85.9	79.2	68.9	60.0	48.2	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.9	38.9	45.9	53.8	60.3	68.0	76.9	78.0	71.3	62.8	53.0	42.2	(29)	
		MCDB	38.6	41.9	52.1	62.3	68.7	71.9	81.4	82.7	75.0	67.1	58.3	45.5	(30)	
	5%	WB	33.6	36.5	43.6	51.1	58.6	66.6	75.1	76.4	69.4	61.0	50.3	39.2	(31)	
		MCDB	36.1	39.6	48.4	58.5	65.0	70.4	79.0	80.3	73.3	65.4	54.9	42.2	(32)	
	10%	WB	31.6	34.5	41.4	49.1	57.1	65.1	73.5	75.1	67.9	59.1	47.8	36.5	(33)	
		MCDB	34.3	37.3	45.4	55.5	62.6	68.5	77.2	78.4	71.8	63.4	52.1	39.5	(34)	
	5% DB	MDBR	14.4	13.6	13.0	14.4	12.1	8.4	7.5	9.4	13.0	15.2	14.2	14.3	(35)	
		MCDBR	15.9	15.9	18.4	22.3	21.7	15.2	13.2	12.9	15.7	17.5	18.3	15.8	(36)	
		MCWBR	12.9	12.3	12.1	11.9	11.1	8.0	7.6	7.5	9.7	11.7	13.4	12.8	(37)	
		MCWBR	14.3	15.0	16.7	19.3	17.0	10.8	11.4	10.5	12.4	14.9	16.7	14.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.357	0.409	0.481	0.551	0.582	0.605	0.561	0.508	0.450	0.444	0.402	0.360	(40)	
		taud	2.123	2.009	1.869	1.768	1.772	1.798	1.983	2.121	2.203	2.081	2.100	2.144	(41)	
	Ebn at Noon		242	246	240	233	230	225	233	243	249	232	223	230	(42)	
		Edh at Noon	34	45	58	68	69	68	56	48	41	41	35	31	(43)	
(44) All-Sky Solar Radiation	RadAvg		704	1009	1398	1624	1676	1583	1427	1438	1334	1073	721	614	(44)	
	RadStd		50	79	89	101	167	161	177	110	130	87	110	44	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page